AMRITA VISHWA VIDYAPEETHAM
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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I132*

Team Name: *JASMIN*

Domain: *6 : Healthcare, Biotechnology Human Well-being*

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Presentation Outline

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Problem Statement

A non-invasive, closed-loop smart patch to safely guide throat movements during swallowing attempts in stroke survivors, preventing fatal choking.

Problem Understanding

- Who experiences this problem : Patients who have survived strokes commonly have been noted to have difficulty swallowing food or liquids. The patient's families and care takers also struggle as they need to be really careful.
- What causes the problem : Stroke or neurodegenerative disorders can affect parts of the brain which is involved in controlling swallowing. Therefore, this makes it difficult for the muscles that are involved in swallowing to work in tandem. In worst cases, food or liquid may sometimes enter the air-way which leads to, Aspiration.
- How does it affect users or stakeholders : Patients usually experience chocking , malnutrition,dehydration, and lung infection.Caregivers/nurse have to constantly monitor them to make sure they swallow safely which becomes stressful for both patients and their families.

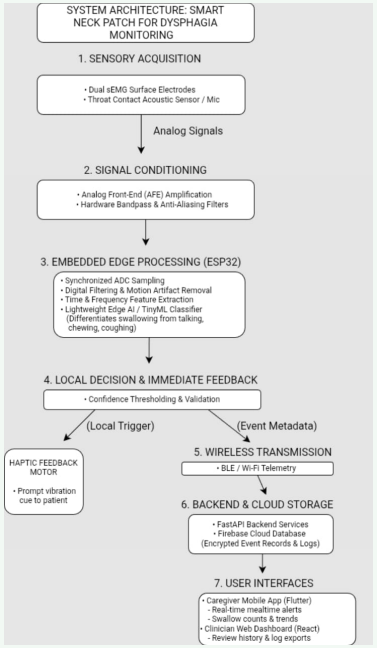
Existing Gaps & Challenges

- Existing gap 1 : Current swallowing assessment methods often require hospital equipment and cannot continuously monitor patients during everyday activities.
- Existing gap 2 : Some existing devices can identify swallowing problems, but they may not alert caregivers in time.
- Main challenge that remains unsolved : Developing an affordable, non-invasive, and wearable system that can reliably detect swallowing patterns and provide timely feedback to improve patient safety and reduce the burden on caregivers.

Our Proposed Solution

- Core idea or product : A smart neck patch designed to help stroke survivors with swallowing difficulties by monitoring their swallowing patterns and providing timely feedback.
- How the solution works at a high level : The patch uses sensors to detect muscle activity and swallowing sounds. An AI system analyses these signals to identify possible swallowing difficulties and provides gentle haptic feedback when needed, while helping caregivers monitor the patient.
- Main value delivered to users : The solution aims to make swallowing safer, reduce the need for constant manual monitoring, and give caregivers better support in caring for stroke survivors.

Technical Approach



Sensors collect signals, AI detects possible issues, and the patch alerts caregivers.

Tech Stack

Frontend: Flutter for caregiver monitoring

Backend: FastAPI for processing data and alerts.

Database: Firebase for storing patient data and records..

AI/ML, APIs, or Hardware: Edge AI, sEMG and acoustic sensors, ESP32, and a vibration motor for detecting swallowing patterns and providing feedback.

Innovation & Uniqueness

- What is new or different : The smart neck patch combines muscle activity and swallowing sound sensors with AI to detect possible swallowing difficulties and provide timely feedback.
- Why is this approach suited to the problem : It is non-invasive, wearable, and designed for continuous monitoring, making it suitable for use at home as well as in healthcare settings.
- What is the key differentiator : Unlike systems that only monitor swallowing, our approach combines multimodal sensing, AI-based detection, and haptic feedback to support patients and caregivers in real time.

Feasibility & Scalability

- **Feasibility:** The prototype can be built using sEMG and acoustic sensors, an ESP32, a vibration motor, and a lightweight AI model. It requires sensor integration, data collection, model training, and testing.
- **Scalability:** The system can be improved for different patients and expanded with a caregiver dashboard, remote monitoring, and personalised detection models.
- **Risks:**Sensor noise, inaccurate detection, and differences between patients may affect reliability. These can be reduced through signal filtering, patient-specific calibration, and clinical testing before real-world use.

Cost for Product (Optional)

- Development or hardware cost : 5,000–10,000 for sensors, ESP32, vibration motor, circuit components, and prototype fabrication.
- Hosting or maintenance cost : 0–2,000 for initial cloud services, data storage, and basic maintenance.
- Estimated total : 5,000–12,000 for an initial prototype.

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: A stroke survivor wears the smart neck patch while eating at home. It monitors swallowing patterns and provides feedback when a possible swallowing difficulty is detected.
- Use case 2: Family members receive alerts when the system detects a possible swallowing issue, helping them respond quickly.
- Use case 3: Doctors can review the recorded data to understand the patient's swallowing patterns and track changes over time.
  Note: Additional slides may be included only for this section.

Impacts and Benefits

- Benefit to the target users: Helps stroke survivors monitor swallowing difficulties and supports caregivers in responding to possible problems.
- Social, economic, or environmental impact : Can reduce the burden on caregivers, support home-based monitoring, and potentially reduce healthcare costs through early detection.
- Expected measurable outcome : A working prototype that detects swallowing patterns, provides timely alerts, and measures detection accuracy and response time during testing.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 3	Target number:3.8	Good Health and Well-being	Promotes better healthcare and supports stroke survivors with swallowing difficulties.
SDG 9	Target number:9.5	Industry, Innovation and Infrastructure	Encourages innovation through wearable sensors and AI-based healthcare technology.

Individual Member Contribution

Member	Contribution or responsibility
MHALSA PADMAKUMAR	Responsible for selecting the sensors, setting up the ESP32, and planning how the neck patch will collect swallowing signals.
SHIZA CKV	Focuses on processing sensor signals and developing an AI model to identify possible swallowing difficulties.
ANWAIKA SANVI P T	Works on the caregiver dashboard, patient records, and alerts to help caregivers monitor swallowing activity.
JOSHIKA CD	Brings the hardware and software together, plans prototype testing, reviews the results, and helps prepare the project documentation and presentation

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